Emerging Technologies Report

Generative AI-Assisted Decision-Support Systems in NHS-Style Healthcare Environments

Name: Lola Beatrice Braut

Programme Area: Digital Technologies

Module Tutor: Kosmas Kosmopoulos

Abstract

Generative artificial intelligence is changing how software systems process information, automate routine tasks, and support human decision-making. One area where these capabilities may offer significant practical value is healthcare, where clinicians spend a considerable proportion of their working day completing documentation and other administrative activities. This paper evaluates several emerging technologies before identifying generative AI as the most appropriate solution for reducing administrative workload within NHS-style healthcare environments.

A conceptual AI-assisted platform is proposed to support clinical documentation, patient triage, workflow automation, and integration with electronic health record systems. The proposed system offers a practical approach to reducing routine administrative tasks while allowing clinicians to retain responsibility for all clinical decisions. Although challenges relating to data protection, regulation, cybersecurity, and AI reliability remain, generative AI has the potential to improve the efficiency of healthcare software when implemented with appropriate safeguards, governance, and human oversight.

Keywords: Generative artificial intelligence, healthcare systems, software development, emerging technologies, intelligent automation, decision-support systems

1 Introduction

Software development has shifted beyond creating applications that simply perform predefined tasks. Modern software increasingly incorporates artificial intelligence, cloud-native computing, connected devices, automation, and data-driven decision-making to support more intelligent, adaptable, and scalable systems. As these technologies continue to mature, organisations are redesigning existing software to improve efficiency, reduce manual processes, and respond more effectively to changing operational demands (Alenezi and Akour, 2025).

Generative artificial intelligence is increasingly being explored as a way of supporting information-heavy and administrative work across different sectors. In healthcare, its potential is closely linked to the large amount of documentation, communication, and record management involved in everyday clinical practice (Sauvola et al., 2024).

Healthcare is one such sector. Although electronic health records and digital platforms have become standard across many organisations, administrative work continues to consume a significant proportion of clinicians' time. Producing clinical notes, updating patient records, reviewing referrals, and completing routine documentation can shorten the time available for direct patient care while contributing to increasing operational pressure (Bajwa et al., 2021). This creates a need for technologies that can reduce administrative burden without introducing new risks to clinical decision-making, patient information, or accountability.

The central question is whether generative AI can reduce administrative workload in NHS-style healthcare without compromising clinical responsibility, security, or oversight.

2 Review of Emerging Technologies in Software Development

Software development now focuses beyond building standalone applications to creating connected systems capable of processing large volumes of information, supporting intelligent decision-making, and adapting to changing user requirements. Advances in cloud computing, artificial intelligence, cybersecurity, connected devices, and automation have changed both the architecture of software systems and the way organisations develop, deploy, and maintain them. Each technology addresses different challenges, making their suitability dependent on the problem being solved (Alenezi and Akour, 2025).

Artificial intelligence is now widely used in software systems for prediction, automation, and information processing. Machine learning models can recognise patterns, analyse data, generate predictions, and automate tasks that previously relied on human judgement. Generative AI has extended these functions by producing natural language responses, summarising information, generating software code, and assisting users with knowledge-based tasks. This has widened the range of activities that software can support, especially where information is unstructured or heavily dependent on language. Such developments have led to wider adoption of AI-powered software across industries where large volumes of information must be processed quickly and accurately (Sauvola et al., 2024).

Cloud-native computing supports modern software applications by providing scalable infrastructure without relying entirely on locally managed servers. Distributed services, containers, and microservices allow applications to respond to changing demand while reducing the operational overhead associated with traditional infrastructure (Kratzke and Quint, 2017).

The Internet of Things (IoT) connects physical devices through software, sensors, and communication networks, allowing information to be collected continuously from real-world environments. Within healthcare, IoT devices are commonly used for patient monitoring, equipment tracking, and environmental sensing, providing data that can support operational decision-making and predictive analytics (Al-Fuqaha et al., 2015).

Cybersecurity remains an essential component of software development as organisations become increasingly dependent on cloud services, connected devices, and digitally integrated systems. Modern security platforms combine encryption, behavioural monitoring, automated threat detection, access controls, and zero-trust principles to reduce the likelihood and impact of cyberattacks while protecting sensitive information. This is paramount in healthcare, where compromised systems or unauthorised access can affect not only data confidentiality but also the reliability and availability of services that clinicians depend on (Rose et al., 2020).

Automation technologies, including robotic process automation (RPA), cut the need for repetitive manual work by completing structured processes such as scheduling, document handling, reporting, and data entry. When integrated with artificial intelligence, these systems can process less structured information and support more complex workflows than traditional rule-based automation (Van der Aalst et al., 2018).

Table 1 shows that no single technology addresses every aspect of the proposed healthcare application. While cloud-native computing, cybersecurity, IoT, and automation each solve specific technical challenges, generative AI offers the broadest combination of capabilities required to reduce clinician administrative workload.

Table 1: Comparison of Emerging Technologies for the Proposed Healthcare System

Technology	Primary Strength	Main Limitation	Suitability
Generative AI	Clinical documentation, decision-support, workflow automation, natural language processing	Requires human verification and careful governance to minimise inaccurate or biased outputs	**High**
Cloud-Native Computing	Scalable infrastructure, system integration, remote accessibility	Dependency on cloud providers and data governance requirements	**High**
Internet of Things (IoT)	Real-time patient monitoring, connected medical devices, operational data collection	Greater cybersecurity exposure and device management complexity	**Medium**
Cybersecurity Technologies	Protection of patient records, access control, regulatory compliance	Supports security rather than reducing administrative workload	**Medium**
Automation and RPA	Efficient automation of repetitive administrative processes	Limited flexibility when handling complex or unstructured clinical information	**Medium**

While several technologies received a high suitability rating for specific functions, none addresses the language-based nature of clinical documentation as directly as generative AI. This makes it the strongest candidate for the proposed healthcare application rather than simply another supporting technology.

3 Benefits, Risks, Characteristics, and Disruptive Trends of Emerging Technologies

Emerging technologies rarely operate in isolation. Few software applications now rely on a single technology. Artificial intelligence, cloud computing, connected devices, automation, and cybersecurity each contribute different capabilities that influence how applications are developed, deployed, and maintained. Rather than replacing traditional software engineering, these technologies extend it by enabling software to process larger volumes of information, automate routine processes, and support more informed decision-making (Alenezi and Akour, 2025). This shift has also changed the software development process. Cloud-native architectures simplify deployment and maintenance, while automation reduces repetitive development and administrative tasks. AI-assisted tools can support testing, code generation, debugging, and documentation. IoT extends software beyond desktop and mobile applications by connecting physical devices with digital services, while cybersecurity has become part of software design from the outset rather than an activity performed after deployment (Souppaya et al., 2022).

The same technologies also introduce technical and organisational risks. AI systems may produce inaccurate or biased outputs if trained using incomplete or unrepresentative data, making human verification essential in healthcare (Dwivedi et al., 2023). This becomes particularly important when AI-generated information is used to support clinical or administrative decisions, as errors may be difficult to identify if the output appears convincing or authoritative. Cloud-native platforms may expose organisations to vendor lock-in, service disruption, or data sovereignty concerns, while connected IoT devices increase the number of potential entry points for cyberattacks (Al-Fuqaha et al., 2015). These risks can also overlap, as healthcare applications increasingly depend on interconnected systems in which cloud services, devices, and AI tools exchange sensitive information. Automation may also require organisations to redesign existing workflows and invest in workforce training as job roles evolve (Van der Aalst et al., 2018). As a result, successful adoption depends not only on the technology itself, but also on how well organisations manage security, oversight, staff readiness, and changes to existing processes.
Not every emerging technology contributes equally to the problem addressed in this report. Generative AI differs because it combines language processing, contextual reasoning, and intelligent automation within a single platform. That makes generative AI particularly suitable for clinical documentation, patient triage, appointment summaries, and decision-support.

Generative AI is becoming a standard feature of commercial software rather than a specialist tool used only by technical teams. AI copilots now assist developers with coding, testing, debugging, and technical documentation, while enterprise platforms have integrated AI directly into everyday workflows. Growing adoption has also increased scrutiny from governments and regulatory bodies, leading to new requirements covering transparency, accountability, copyright, data protection, and responsible AI use. Future software projects will need to satisfy these expectations alongside technical and business requirements (Sauvola et al., 2024).

The comparison presented in this section identifies generative AI as the most suitable technology for the proposed healthcare application. Unlike the other technologies reviewed, it directly addresses the administrative and communication challenges faced by clinicians through natural language processing, structured documentation, and information management. These characteristics make it well suited to healthcare environments where efficient documentation and timely access to information are essential.

4 Disruptive Potential of Emerging Technologies

The influence of emerging technologies can be seen in the way organisations now develop software, manage information, and deliver digital services. Many activities that previously relied on manual processes are now completed through intelligent software, changing established workflows and altering expectations for both developers and end users. As a result, software is expected to perform a wider range of tasks, respond more quickly to changing requirements, and integrate with other digital systems as part of everyday operations (Alenezi and Akour, 2025).

Software engineering has changed alongside these expectations. AI-assisted coding tools, automated testing, continuous integration pipelines, and cloud-based development environments shorten the time required to build, test, and maintain applications. This has accelerated software release cycles and allowed organisations to introduce new features and respond to user feedback more quickly (Sauvola et al., 2024).

The impact extends beyond software development. Organisations have redesigned many routine activities around digital workflows, reducing dependence on paper-based records, manual administration, and disconnected information systems. Users have also adapted their expectations, with features such as conversational interfaces, intelligent search, automated recommendations, and real-time information becoming common across many applications. Software that cannot provide these functions is more likely to be viewed as inefficient or outdated (Verhoef et al., 2021).

Healthcare illustrates this change particularly well. Electronic health records, cloud-based platforms, and connected medical devices have improved access to patient information and shortened many manual processes. Administrative work, however, remains a major source of pressure for clinicians. Preparing consultation notes, updating patient records, processing referrals, and retrieving information continue to consume valuable clinical time. This creates a particular problem because much of the work involves interpreting, summarising, or producing unstructured written information rather than simply completing a predefined automated task. Generative AI addresses this challenge more directly than the other technologies reviewed by assisting with language-based tasks that cannot be solved through infrastructure, security, or automation alone (Bajwa et al., 2021).

Disruptive technologies also introduce new responsibilities alongside new opportunities. Organisations adopting AI-assisted systems must protect sensitive information, comply with legal and regulatory requirements, monitor the quality of automated outputs, and ensure that important decisions remain subject to human oversight. Workforce training and clear governance are therefore essential if emerging technologies are to be integrated safely into existing systems and working practices (Mennella et al., 2024).

5 Evaluation and Selection of Generative Artificial Intelligence

Several emerging technologies were considered for the proposed healthcare application, including cloud-native computing, the Internet of Things, cybersecurity technologies, automation, and generative artificial intelligence. Each technology addresses a different requirement, but generative AI is the closest match to the main problem identified in this report: reducing the administrative workload linked to clinical documentation and information management.

Healthcare professionals spend considerable time producing consultation notes, reviewing patient records, completing referral documentation, and retrieving information from different systems. Much of this work depends on language. Generative AI is suited to this type of task because it can summarise text, generate structured

documentation, retrieve relevant information, and present draft outputs for clinician review (Bajwa et al., 2021).
For the purposes of this report, the primary end-user group consists of NHS general practitioners (GPs). GPs routinely produce consultation notes, referrals, and patient records while managing high patient volumes, making administrative workload a suitable problem for AI-assisted documentation. The proposed platform could be adapted for other healthcare professionals and the design is primarily evaluated from the perspective of NHS GPs.

Figure 1: Technology Evaluation and Selection Matrix. Author’s own work.

Figure 1 summarises the technology evaluation process. Generative AI was identified as the most suitable option because it directly addresses the language-intensive administrative activities associated with clinical documentation and information retrieval. Other technologies, including cloud-native computing, IoT, cybersecurity, and automation, remain important components of the proposed solution by providing infrastructure, security, and supporting capabilities; however, they do not directly address the primary challenge of reducing documentation workload for clinicians.

The selection of generative AI does not remove the need for clinical judgement. AI-generated content may be inaccurate or incomplete, particularly when source data are limited or ambiguous. Outputs generated by the proposed system should be reviewed and approved by a healthcare professional before becoming part of the patient’s record. This human review helps reduce the risk of incorrect documentation while ensuring that responsibility for clinical decisions remains with qualified practitioners. The proposed system is intended to support clinical practice rather than replace professional judgement, and so this balance between automation and clinician oversight reflects current guidance on the responsible use of AI in healthcare (Mennella et al., 2024).

6 Proposed AI-Assisted Healthcare System Architecture

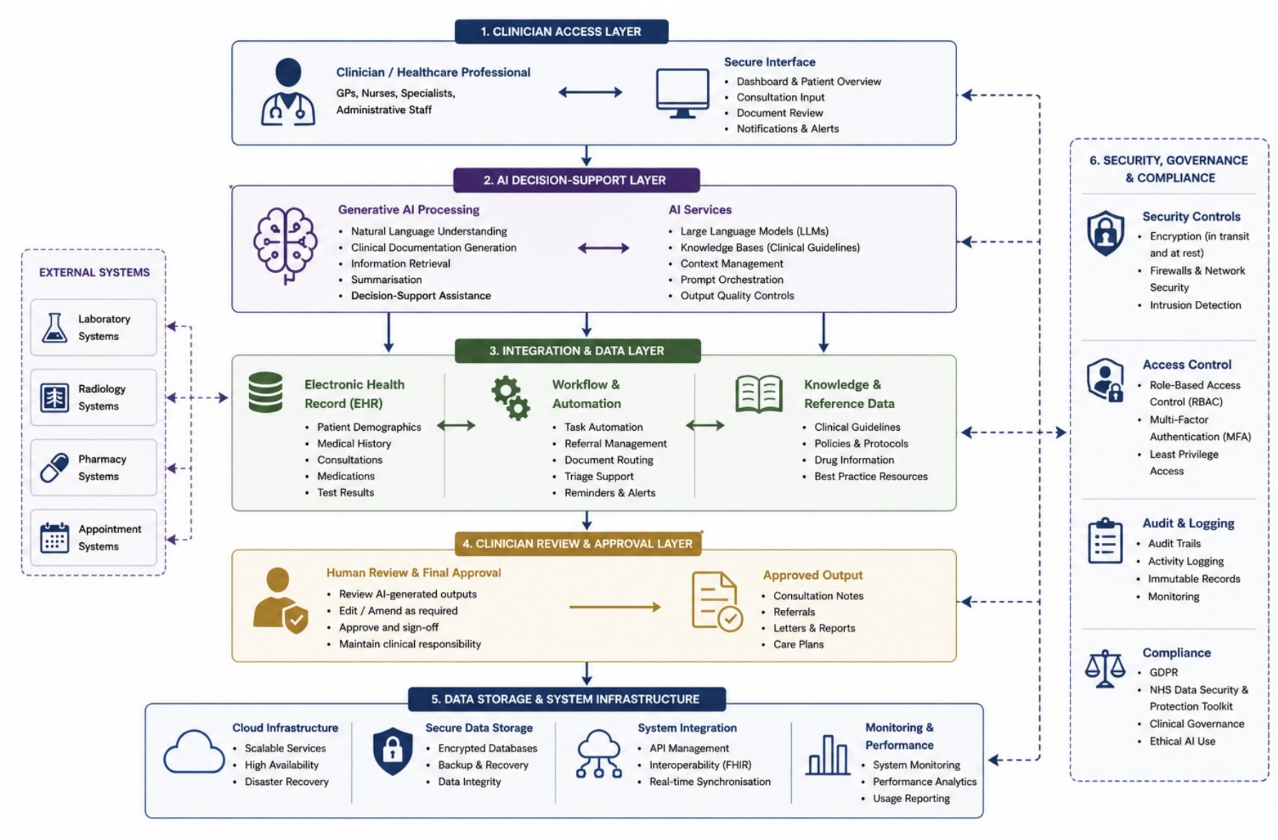

Figure 2: Conceptual Architecture of the AI-assisted Healthcare System. Author's own work.

Figure 2 illustrates the conceptual architecture of the proposed AI-assisted healthcare system. It shows how generative AI, electronic health record integration, workflow automation, and clinician verification operate as a single software solution.

Clinicians access the platform through a secure interface where consultation notes, patient records, referrals, and appointment information are submitted for processing. The AI layer generates draft clinical documentation, retrieves relevant patient information, and prepares outputs for clinician review before any information is approved. This review stage ensures that clinical responsibility remains with the clinician rather than the AI system.

Information moves through the platform in a structured sequence. During or following a patient consultation, the clinician's input is combined with relevant information retrieved from the electronic health record before being processed by the AI layer. The system generates draft consultation notes, referral summaries, or administrative responses that can be edited where necessary before final approval. Only clinician-approved information is written back to the electronic health record, making sure that AI assists documentation without replacing clinical decision-making. The effectiveness of the platform also depends on the quality of the information available within existing electronic health record systems. Incomplete, inconsistent, or outdated patient records may reduce the quality of AI-generated outputs, reinforcing the need for clinician review throughout the process (Bajwa et al., 2021).

The platform is supported by cloud infrastructure, encrypted databases, role-based access control, audit logging, and cybersecurity controls to protect sensitive patient information and support regulatory compliance. The architecture can be deployed across multiple departments while maintaining secure access to shared healthcare records. The modular design also supports future expansion. Additional services, such as predictive analytics, clinical decision-support models, or integration with wearable medical devices, could be incorporated without fundamentally changing the architecture.

6.1 End-user Feedback and Prototype Refinement

The initial prototype would be evaluated by a small group of NHS general practitioners using realistic consultation scenarios. Feedback could be gathered through short questionnaires, observation during task completion, and semi-structured interviews focusing on usability, documentation quality, workflow integration, and trust in AI-generated outputs.
Early feedback would be expected to identify several areas for improvement. Users may request clearer separation between AI-generated text and clinician-entered information, improved editing tools, and more visible indicators showing when documentation has been reviewed and approved.
Following this feedback, a second iteration of the prototype would include highlighted AI-generated sections, simplified editing controls, and a more prominent clinician approval stage before information is written back to the electronic health record. These changes would improve usability while reinforcing clinician oversight and reducing the risk of incorrect documentation being accepted without review.

Table 2 summarises the two iterations of the proposed prototype. The second version incorporates improvements based on expected feedback from NHS general practitioners, demonstrating how user evaluation can refine the design before implementation.

Table 2: Prototype Iterations Based on End-user Feedback

Prototype	**Features**	**Improvements**
Version 1	AI-generated consultation notes, electronic health record integration, and clinician approval before information is saved.	Initial prototype developed to demonstrate the core workflow and AI-assisted documentation process.
Version 2	Improved editing interface, clearer identification of AI-generated content, and a more prominent clinician approval stage.	Refined following feedback from NHS GP users to improve usability, increase transparency, and support safer review of AI-generated documentation.

These refinements improve usability while supporting safer use of AI-generated documentation.

7 Impact of the Proposed System on Healthcare Organisations and Users

The proposed AI-assisted healthcare system is expected to influence both clinical practice and organisational performance by improving clinical workflows and access to patient information. The primary objective is to decrease clinician workload, but the effects extend to patients, healthcare organisations, and day-to-day working practices. For healthcare professionals, the greatest benefit is likely to be a reduction in administrative tasks. Generating draft consultation notes, organising referral information, and retrieving patient records can be completed more quickly with AI assistance, leaving clinicians with more time for direct patient care. Patients may also benefit from shorter waiting times, faster processing of referrals, and better continuity of care as information moves more efficiently between departments (Topol, 2019).

Healthcare organisations may also experience improvements in resource management. Integrating AI with electronic health records and existing digital systems can cut duplicated work, improve communication between departments, and make clinical information easier to access. Eventually, this may improve productivity while helping organisations manage increasing demand for healthcare services (Verhoef et al., 2021).

The system also introduces challenges that must be managed carefully. AI-generated content may contain errors or incomplete information, making clinician review essential before information is added to a patient's record. Healthcare providers must also protect sensitive data through encryption, role-based access control, audit logging, and compliance with data protection legislation. Staff training is equally important to make sure the technology is used safely and consistently (Mennella et al., 2024). These changes would require clinicians to adapt existing documentation workflows, but they may also cut the amount of time spent completing administrative tasks during and after consultations.

Introducing an AI-assisted platform requires investment in software integration, cloud services, cybersecurity, and workforce training. Initial implementation costs may be high; however, the decrease in administrative workload and more efficient use of clinical time may provide long-term value. Successful adoption would ultimately depend on balancing technical capability with patient safety, effective governance, and continued human oversight.

8 Ethical, Legal, Social, and Economic Challenges of Emerging Technologies

Emerging technologies have transformed software development across many sectors, but their adoption also creates ethical, legal, social, and economic challenges. Organisations implementing technologies such as artificial intelligence, cloud computing, automation, and the Internet of Things must consider issues including fairness, accountability, data privacy, cybersecurity, workforce adaptation, regulatory compliance, and implementation costs. The importance of these challenges varies depending on the application, but they become particularly significant in healthcare because software systems may influence patient safety, confidentiality, and public trust (Mennella et al., 2024).

Table 3: Ethical, Legal, Social, Economic, and Cybersecurity Challenges for the Proposed AI-Assisted Healthcare System

Area	Challenge for the Proposed System	Mitigation
Ethical	AI-generated documentation or recommendations may contain bias or inaccuracies, potentially affecting clinical judgement and patient outcomes.	Maintain clinician approval for all AI-generated outputs, regularly audit system performance, and use representative training data.
Legal	Patient information must comply with GDPR, NHS data governance requirements, and professional accountability standards.	Apply encryption, role-based access control, audit logs, and documented governance procedures.
Social	Healthcare professionals may be reluctant to adopt AI-assisted systems because of concerns about trust, professional autonomy, and changing job roles.	Provide staff training, phased implementation, and clear guidance that AI supports rather than replaces clinical judgement.
Economic	System integration, cloud infrastructure, cybersecurity, and staff training require significant investment before operational benefits are realised.	Introduce the platform in phases and evaluate long-term savings against implementation and maintenance costs.
Cybersecurity	Sensitive patient records may become targets for cyberattacks, unauthorised access, or ransomware incidents.	Implement continuous monitoring, penetration testing, encryption, multifactor authentication, and regular security updates.

The challenges summarised in Table 3 do not prevent the adoption of generative AI, but they demonstrate that successful implementation depends on more than technical performance. Ethical governance, legal compliance, cybersecurity, and workforce engagement are equally important to ensure the technology is used safely and responsibly (Floridi and Cowls, 2022; Souppaya et al., 2022). Within healthcare, AI should complement clinical practice by assisting with administrative tasks while leaving responsibility for patient care and clinical decision-making with qualified healthcare professionals (Floridi and Cowls, 2022).

The development of generative AI creates an additional regulatory challenge because technical capabilities may change faster than organisational governance and legislation. Compliance therefore cannot be treated as a one-time activity completed when a system is introduced. Changes to AI models, training data, or system functionality may alter the reliability of outputs and introduce risks that were not identified during the original assessment. Healthcare organisations would need continuous monitoring, documented system updates, periodic reassessment, and clear procedures for responding to changes in performance or regulatory requirements.

Human oversight also raises questions about accountability when AI-generated information contributes to an error. Although requiring clinicians to review and approve generated documentation provides an important safeguard, responsibility may become less straightforward when an incorrect output has influenced the final decision. Accountability could involve the clinician who approved the information, the healthcare organisation that implemented the system, and the technology provider responsible for the AI model. Clear governance would be needed to define these responsibilities rather than assuming that clinician approval alone resolves the issue.

The challenges identified in this section do not outweigh the potential benefits of the proposed system. Lessening administrative workload, improving access to patient information, and streamlining clinical documentation may increase efficiency, provided that strong governance, continuous monitoring, and human oversight are maintained. Careful implementation enables healthcare organisations to benefit from emerging technologies while managing the ethical, legal, social, economic, and cybersecurity risks associated with their adoption (Topol, 2019).

9 Future Improvements and Recommendations

The proposed AI-assisted healthcare system provides a practical approach to reducing clinician administrative workload, but several improvements could increase its effectiveness as healthcare technology continues to develop.

One priority should be improving transparency. Future versions could incorporate explainable AI so that clinicians can better understand how recommendations and generated documentation are produced. Greater transparency would improve trust and make it easier to identify incorrect or biased outputs (Arrieta et al., 2020).
Better integration with existing healthcare software is another important area for development. Stronger interoperability between electronic health record systems, laboratories, pharmacies, and other healthcare providers would lessen duplicated work and improve the flow of clinical information between departments.
Future releases could also include predictive analytics to identify patient admission trends, highlight patients requiring earlier intervention, and support resource planning. Alongside these improvements, cybersecurity measures should continue evolving to protect sensitive healthcare information as digital systems become more interconnected (Topol, 2019).

Successful adoption also depends on people rather than technology alone. Regular staff training, system evaluation, and updates to governance policies will help ensure that AI continues to support clinicians safely and effectively while remaining compliant with future legal and regulatory requirements (Mennella et al., 2024). Continued investment in these areas would allow the system to develop in conjunction with changing healthcare needs while maintaining patient safety, regulatory compliance, and clinical oversight.

10 Conclusion

The technologies reviewed in this report contribute to healthcare software in different ways, but generative artificial intelligence offers the strongest match for reducing clinician administrative workload within NHS-style healthcare environments. Cloud computing, automation, cybersecurity, and the Internet of Things each address different technical requirements, and their greatest value is often realised when they are used together rather than separately. This report evaluated several emerging technologies before identifying generative artificial intelligence as the most appropriate choice for reducing clinician administrative workload within NHS-style healthcare environments. Its ability to support clinical documentation and information management made it the most suitable technology for the proposed healthcare application.

A conceptual AI-assisted healthcare platform was proposed to demonstrate how generative AI could be integrated with electronic health records, workflow automation, and clinician verification. Maintaining human oversight throughout the process ensures that AI remains a decision-support tool rather than a replacement for professional judgement. The report also recognised that successful adoption depends on more than technical performance. Ethical, legal, social, economic, and cybersecurity considerations must all be addressed to protect patient information, maintain public trust, and ensure compliance with healthcare regulations.

When implemented responsibly, generative AI offers a practical opportunity to curtail administrative pressure, improve access to clinical information, and give healthcare professionals more time to focus on patient care. The conceptual platform presented in this report demonstrates how generative AI can be integrated into healthcare software to address a practical challenge within healthcare administration. Continued advances in AI, combined with responsible implementation and clinician oversight, are likely to strengthen the role of intelligent software within future healthcare systems.

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